

Nuclear Shell Regime Audit: The Magic-Number Shift Theorem and the Discovery of the N=14 Statistical Rupture

—The η Hierarchy Structure, Bai-Perron Optimal 10 Breakpoints from Zero-Preset Exhaustive Search, and Six Quantitative Theorems

Shuiping Tang

Independent Researcher

Abstract

The central axiom of the nuclear shell model—that nucleons move independently in a static mean field—has never been systematically questioned. This paper deploys the Factor Hierarchy Law to nuclear physics for the first time, performing a zero-preset regime audit on the full AME2020 nuclide dataset—the algorithm does not know of the existence of "magic numbers" or "shells." Exhaustively testing 18 candidate execution factors, the optimal combination under the $VIF < 5$ constraint is $\text{mass_excess_keV} + \text{beta_decay_keV} + N_over_Z$ (adjusted $R^2 = 0.8052$), with Bayesian model averaging confirming its statistical irreplaceability with a weight of 1.0000. The core findings unfold in four tiers. Tier 1: the η hierarchy—the ratio of interaction effect to direct effect for continuous N/Z exceeds 10^6 ($\eta > 10^6$), that for the binary magic-number marker is $\eta = 19.19$, and that for the continuous shell-correction energy is $\eta \approx 0$. This three-tier η hierarchy points to a single conclusion: under continuous-variable proxies, N and Z behave as regime factors (function realized through interaction effects), the shell correction is an execution factor (directly explaining binding energy), and independent particle motion exists but is not the rule-maker. Tier 2: Triple cross-validation and supplementary evidence for the $N=14$ statistical rupture—all three inv-family activation operators lock onto $N=14$ ($F = 415\text{--}572$), the threshold interaction model locks onto $N=25$ as the optimal threshold, the PICO intervention effect attains its maximal intercept jump at $N=14$ (-245.98 keV, $p < 0.001$), and all eight breakpoints of the kernel change-point cosine kernel concentrate at $N < 15$. The existence of a statistically significant structural rupture at $N=14$ is a statistical fact confirmed by three independent tools. Tier 3: The full- N -axis Bai-Perron sequential blind test locks onto an optimal 10 regime-switch breakpoints with overwhelming statistical evidence ($BIC = 13793.7$). The individual ΔBIC values of all five traditional magic numbers conform to the Statistical Distinction Theorem of Shell-Closure Physical Mechanisms— $N=28$ is of deformation-transition type ($\Delta BIC = 415.9$), and $N=20$ is of single-particle-gap type ($\Delta BIC = -514.4$). Magic numbers are not optimal partition points in the statistical sense—the Magic-Number Shift Theorem, the Statistical Distinction Theorem, and the Magic-Number Node Theorem jointly establish the repositioning of traditional magic numbers within the statistical structure. Tier 4: Three lines of negative control exclude statistical artifacts (permutation test $p = 0.002$, fake-data $R^2 = 0.0007$, WLS residual Chow $F = 482.55$). This paper refines six quantitative theorems, establishes the statistical existence of regime-switch signals in the nuclear shell, and constitutes the necessary premise for the subsequent argument that "independent particle motion is not the fundamental organizing principle of the nuclear shell."

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1. Introduction

1.1 The Axiom of the Shell Model and the Question Never Asked

The nuclear shell model is the cornerstone of nuclear physics. Its central axiom is as simple as it is powerful: nucleons move independently in a static mean field (Mayer & Jensen, 1955). The direct corollary of this axiom is the existence of magic numbers—when large energy gaps appear between single-particle levels, nuclei exhibit extra stability. $N = 2, 8, 20, 28, 50, 82, 126$ —these numbers constitute the fundamental coordinates of the nuclear shell.

However, independent particle motion is not exact. Nucleons interact through complex many-body correlations—pairing, quadrupole deformation, tensor forces—effects that lie beyond the descriptive power of a static mean field. To remedy this deficiency, the shell-correction method (the Strutinsky method) was introduced: it is a phenomenological compensation for the discrepancy between independent particle motion and experiment (Strutinsky, 1967).

This theoretical framework has achieved enormous success. Yet beneath that success, a fundamental question has never been systematically interrogated: Is independent particle motion itself the fundamental organizing principle of the nuclear shell? Or is it merely the approximate manifestation of a deeper structure in the spherical-symmetry limit?

This question differs from "Is the shell model accurate?"—everyone knows the shell model requires corrections. This question probes a deeper logic: If independent particle motion is the fundamental organizing principle, why does it require such a massive external compensation to agree with experiment? If it is only an approximation, what is the true organizing principle?

1.2 The Factor Hierarchy Law and Testability Norms

The Factor Hierarchy Law provides a novel methodological framework for answering this question. The law states: In any complex system driven by multiple quantifiable factors, factors naturally divide into two hierarchical levels—regime factors and execution factors. Regime factors determine which set of rules currently applies; their function is manifested not in direct explanatory power but in the interaction effects through which

they modulate the exposure coefficients of execution factors. When a regime factor crosses a critical value, the factor weights of the entire system are systematically reset (Tang, 2026g, 2026r).

The law comprises three axioms:

- **Axiom 1 (Factor Dichotomy):** In any complex system driven by multiple quantifiable factors, factors naturally divide into two hierarchical levels: regime factors and execution factors.
- **Axiom 2 (Regime Switch):** When a regime factor crosses a critical value, the factor weights of the entire system are systematically reset.
- **Axiom 3 (Interaction Effect):** The function of a regime factor is manifested not in direct explanatory power but in the interaction effects through which it modulates the exposure coefficients of execution factors.

The Factor Hierarchy Law is not a theoretical hypothesis—it is an empirical regularity that emerges automatically from objective data. Its discovery began in finance (Tang, 2026a-f) and subsequently obtained independent cross-disciplinary verification in astronomy (Tang, 2026h-j), statistical physics (Tang, 2026k-m), particle physics (Tang, 2026n-q), and chemistry (Tang, 2026s-u). Nuclear physics is the sixth discipline to be deployed.

The operational implementation of the Factor Hierarchy Law is the Testability Norms—comprising six functional dimensions: search completeness, modulation relationship, structural breakpoints, causal direction, problem completeness, and process closure. The Testability Norms are not bound to any specific tools—the functional dimensions are eternal, and the implementation tools are variable (Tang, 2026g, 2026r). These norms, together with Popperian falsifiability, constitute the complete chain of scientific methodology: Popper's falsifiability guards the entrance to science, and testability guards the exit of scientific discovery.

The above cross-disciplinary verification provides threefold support for the nuclear-physics deployment in this paper. First, the statistical signature of a regime factor—interaction effects vastly exceeding direct effects—has been independently verified in finance and particle physics, ruling out the alternative explanation that "this is an accidental phenomenon peculiar to nuclear physics data." Second, precursor cascades have been precisely and quantitatively verified in both particle physics and chemistry, providing a cross-disciplinary pattern basis for the physical interpretation of $N=14$. Third, activation-operator stratification exhibits the isomorphic behavior of "different operators selectively illuminating different regime centers" in both particle physics and chemistry, making the stratification in nuclear physics yet another instance of a cross-disciplinary universal regularity.

1.3 Historical Record of Cross-Disciplinary Verification

In finance, the China-U.S. interest-rate spread was confirmed as a regime factor—its direct incremental R^2 is near zero, but it is highly significant in the interaction effects modulating the exposure coefficients of VIX and DXY. When the spread crosses the zero axis, the global asset-pricing equation undergoes a structural rupture (Tang, 2026a-f).

In astronomy, stellar spectral type acts as a regime factor, triggering a regime switch in the transit-depth equation at a temperature of 4762 K—the Tang Break, with structural ruptures across all five independent physical dimensions (Tang, 2026h-j).

In statistical physics, the precise information-theoretic duality between the Factor Hierarchy Law and the Ehrenfest phase-transition classification was demonstrated through an analytic mathematical proof on the Onsager exact solution of the 2D Ising model, with a deviation strictly equal to zero (Tang, 2026k). On the 3D Ising model—a rigorously proven NP-complete problem without an analytic solution—hidden structures never before revealed by traditional finite-size scaling analysis were discovered: double-peak fission, transition regimes, and precursor boundaries (Tang, 2026l). The reduction of universality was completed across five phase-transition models, establishing regime boundaries as more fundamental physical quantities than universality classes in finite systems (Tang, 2026m).

In particle physics, color charge was confirmed as a meta-regime factor. The algorithm automatically locked onto the positions of Λ_c^+ , $\Upsilon(1S)$, and $\Upsilon(2S)$ with 10^{-4} GeV precision—without any knowledge of the physical meaning of these particles. The inevitable emergence of $SU(3) \times SU(2) \times U(1)$ was derived from regime boundaries, completing the demotion of gauge field theory (Tang, 2026n-q).

In chemistry, period and valence electrons were confirmed as orthogonal regime factors. The Calcium-Lanthanum Singularity Theorem revealed a quantitative correspondence between regime-switch intensity and orbital angular momentum— $F(l) \propto l^2$. The Chemical Gauge Group Theorem rigorously derived the necessity of s, p, d, and f orbitals from eight mutually exclusive regime states via $SO(3)$ group theory, completing the demotion of the periodic law (Tang, 2026s-u).

1.4 Core Tasks and Argument Boundary of This Paper

This paper deploys the Factor Hierarchy Law to the domain of atomic nuclear physics for the first time. It has four core tasks:

1. To perform a zero-preset regime audit on the full AME2020 nuclide dataset—the algorithm knows nothing of "magic numbers," "shells," or "spin-orbit coupling";
2. To test whether regime-switch centers exist on the N-axis of the nuclear binding-energy equation, and if so, at what positions, and whether they correspond to traditional magic numbers;
3. To test whether N and Z satisfy the defining characteristics of regime factors—interaction effects vastly exceeding direct effects, with negligible direct contribution;
4. To provide the first two key pieces of statistical evidence for the subsequent argument that "independent particle motion is not the fundamental organizing principle of the nuclear shell."

This paper presupposes no nuclear physics theory. The algorithm only automatically detects regime boundaries from the pure data matrix.

It is necessary to clarify that this paper strictly distinguishes the argument boundaries of two propositions:

- **Proposition A (completed in this paper):** The η hierarchy reveals that, under continuous-variable proxies, N and Z behave as regime factors and the shell correction is an execution factor; the Bai-Perron blind test proves that magic numbers are not optimal partition points in the statistical sense; independent particle motion exists but is not the rule-maker.

- **Proposition B (deferred to subsequent papers, Tang, 2026w, 2026x):** Independent particle motion is not the fundamental organizing principle of the nuclear shell—this requires the proof that the regime-switch network systematically outperforms the shell model in completeness, self-consistency, and explanatory power (three lines of evidence: 15% coverage, liquid-drop masking effect, shell-correction compensation traces), jointly supported by the methodological-priority argument.

Unlike the completed demotion arguments in particle physics and chemistry, the regime audit of this paper stops at statistical discovery—the complete argument for the physical proposition is deferred to subsequent papers. This is dictated by the positioning of the first paper in the nuclear-physics series, not by any limitation of argumentative capacity. The task of this paper is to provide the first link—it is the most fundamental link, because if η were of the same order of magnitude as the direct effect, and if Bai-Perron had precisely locked onto all traditional magic numbers, then the entire chain of argument would have no reason to unfold.

2. Data and Methods

2.1 Data Sources

All data come from the international Atomic Mass Evaluation AME2020 and the nuclear properties database NUBASE2020 (Wang et al., 2021). They were obtained through the official channel of the IAEA Atomic Mass Data Center (AMDC), with six files downloaded: `mass_1.mas20.txt`, `massround.mas20.txt`, `rct1.mas20.txt`, `rct2_1.mas20.txt`, `nubase_4.mas20.txt`, and `covariance.mas20`. The data are completely public, and any researcher can independently reproduce all results.

After line-by-line parsing and multiple cross-validations, a master data table of 3558 ground-state nuclides $\times$ 41 columns was ultimately constructed. Ground-state nuclides were identified by a blank space in the 17th column (the `isomer_type` field) of the NUBASE file. The coverage range is: proton number $Z = 0\text{--}118$, neutron number $N = 0\text{--}177$, mass number $A = 1\text{--}295$. The explained variable is `bind_per_A_keV` (binding energy per nucleon), with a mean of 7945.76 keV, a range of $[-2267.0, 8794.6]$ keV, and a standard deviation of 740.9 keV. Nuclear property statistics: 3373 nuclides have known spin-parity, 253 are stable nuclides, 12 are doubly magic, and there are 0 duplicate (Z, N) pairs.

2.2 Candidate Factor Pool

The construction of the candidate factor pool follows the core requirement of search completeness—all candidate factors must be systematically traversed, without selecting factors based on the researcher's prior preferences.

Candidate execution factors (18), covering six categories:

- Mass-related quantities: `mass_excess_keV`, `atomic_mass_micro_u`
- Separation energies: `S2n_keV`, `S2p_keV`, `Sn_keV`, `Sp_keV`
- Decay energies: `Qa_keV`, `QB_keV`, `Q2B_keV`, `beta_decay_keV`
- Nuclear properties: `spin_value`, `parity`, `half_life_sec`
- Composition quantities: `N_minus_Z`, `N_over_Z`, `A_minus_2Z`
- Difference-derived quantities: `dS2n`, `dS2p`

Candidate regime factors (10): `N`, `Z`, `A`, `is_magic_N`, `is_magic_Z`, `is_double_magic`, `N_minus_Z`, `N_over_Z`, `parity`.

Table 1 reports the missing rate of all candidate execution factors. The complete sample size is 1368 nuclides (all 18 factors non-missing). Missing separation energies mainly occur near the drip lines—for these nuclides, two-neutron or two-proton separation energies do not exist, which is a physical fact rather than a data error. The missing rate for `spin_value` is 45.2%, mainly due to a lack of experimental data in the heavy-nucleus region.

Table 1: Summary of Missing Rates for Candidate Execution Factors

Factor	Missing Count	Missing Rate	Reason for Missing
<code>mass_excess_keV</code>	0	0.0%	—
<code>atomic_mass_micro_u</code>	0	0.0%	—
<code>S2n_keV</code>	237	6.7%	No two-neutron separation energy near drip lines
<code>S2p_keV</code>	355	10.0%	No two-proton separation energy near drip lines
<code>Sn_keV</code>	118	3.3%	Lightest nuclides have no single-neutron separation energy
<code>Sp_keV</code>	178	5.0%	Lightest nuclides have no single-proton separation energy
<code>Qa_keV</code>	144	4.0%	Alpha decay channel unavailable for light nuclei
<code>QB_keV</code>	413	11.6%	No beta decay near beta-stability line
<code>Q2B_keV</code>	588	16.5%	Double beta decay only for a few nuclides
<code>beta_decay_keV</code>	295	8.3%	Stable nuclides have no beta decay
<code>spin_value</code>	1609	45.2%	Lack of experimental data in heavy-nucleus region

Factor	Missing Count	Missing Rate	Reason for Missing
parity	185	5.2%	Parity unmeasured for a few nuclides
half_life_sec	471	13.2%	Half-life infinite for stable nuclides
N_minus_Z	0	0.0%	—
N_over_Z	1	0.0%	Denominator zero for Z=0 (neutron)
A_minus_2Z	0	0.0%	—
dS2n	355	10.0%	Difference-derived quantity, first element missing
dS2p	533	15.0%	Difference-derived quantity, first element missing

2.3 Execution Plan for the Testability Norms

This paper strictly follows the six functional dimensions of the Testability Norms. **Table 0** lists the tool selection and rationale for each dimension.

Table 0: Tool Layer of the Six Functional Dimensions and Selections in This Paper

Functional Dimension	Optional Tools (non-exhaustive)	Selection in This Paper	Rationale for Selection
Search Completeness	Exhaustive testing, L1 regularization, random forest importance, Bayesian model averaging, genetic algorithms, mutual information	Exhaustive + L1 + RF + MI + BMA	18 candidate factors, manageable computational load for exhaustive search; BMA quantifies model uncertainty
Modulation Relationship	Interaction effect test, threshold interaction model, local weighted regression, hierarchical regression variance decomposition	Interaction effect test + threshold interaction model	Linear interaction effect directly tests defining characteristics of regime factors; threshold interaction searches for optimal threshold
Structural Breakpoints	Chow test, Bai-Perron sequential test, SupF test, CUSUM test, MOSUM test, PELT algorithm	Chow + SupF + CUSUM + Bai-Perron	Chow for known breakpoint verification, SupF for uncertain breakpoint scanning, CUSUM for parameter drift detection, Bai-Perron for exhaustive multi-breakpoint search
Causal Direction	Granger causality test, instrumental variable method, regression discontinuity design,	Conditional independence + RDD + IV	Cross-sectional data suitable for conditional independence; IV excludes endogeneity; RDD estimates local causal effect

Functional Dimension	Optional Tools (non-exhaustive)	Selection in This Paper	Rationale for Selection
	conditional independence test, DAG		
Problem Completeness	5W2H framework, PRISMA framework, PICO framework, SWOT framework	5W2H (this paper) + PRISMA (Tang, 2026x) + SWOT (Tang, 2026y)	The core task of this paper is initial deployment and blind discovery
Process Closure	PDCA cycle, OODA cycle, DMAIC Six Sigma	PDCA	Complete closed-loop management from audit design to conclusion output

The specific execution plan for each dimension is as follows:

Search Completeness: Exhaustive testing of all 1- to 3-factor combinations of the 18 candidate execution factors is performed, with the optimal combination automatically winning under the dual hard constraints of $VIF < 5$ and $VIF < 10$, sorted by adjusted R^2 . Simultaneously, L1 regularization (LassoCV, 5-fold cross-validation), random forest importance (100 trees), mutual information maximization (fixed random state), and Bayesian model averaging (BIC weights) are deployed to cover both linear and nonlinear dependence relationships.

Modulation Relationship: Interaction effect tests are performed one by one on the 10 candidate regime factors. The regression equation is $\text{bind_per_A_keV} \sim \text{execution factors} + \text{regime factor} + \text{execution factors} \times \text{regime factor}$. A significant interaction term ($p < 0.05$) is a necessary condition for a regime factor. Simultaneously, a threshold interaction model is deployed—searching the N-axis for the optimal threshold value that maximizes the difference in interaction effects.

Structural Breakpoints: The Chow test (whether the equation structure differs at known candidate breakpoints), the Bai-Perron sequential test (exhaustive search for multiple breakpoints, BIC selecting the optimal number of breakpoints), the SupF test (whether any structural rupture exists when the breakpoint location is uncertain), and the CUSUM test (whether equation parameters are drifting) are applied in sequence, covering the complete logical chain from "known breakpoints $\rightarrow$ unknown breakpoints $\rightarrow$ no preset breakpoints." Simultaneously, seven activation operators—raw, inv, log, sqrt, sq, cube, inv_sq—are deployed to detect regime-switch signals under different mathematical transformations.

Causal Direction: Conditional independence tests (whether the regime factor remains significantly correlated with the target after conditioning on execution factors), regression discontinuity design (estimating the local causal effect near a regime boundary), and the instrumental variable method (using A as an instrument for N_over_Z to exclude endogeneity) are deployed.

Negative Controls: Three lines of defense are deployed—a permutation test (500 random shuffles of N labels, recomputing the optimal Chow F), a fake-data control (a pure-noise system without regime switches),

and a weighted least squares (WLS) residual Chow test (using AME2020 official measurement errors as weights), progressing from the weakest to the strongest alternative explanation.

Problem Completeness and Process Closure: This paper covers all seven dimensions of inquiry of the 5W2H framework (What-Who-When-Where-Why-How-How Much). The research process follows the PDCA cycle—Plan (design the audit scheme) → Do (execute the six-dimension tests) → Check (exclude alternative explanations through negative controls) → Act (output the regime identification results).

2.4 Methodological Calibration Foundation

The core statistical tools used in this paper—the Chow test and its derivatives (Bai-Perron sequential test, SupF test, CUSUM test)—have passed rigorous econometric calibration on the Onsager exact solution of the 2D Ising model (Tang, 2026k). The core conclusions of this calibration include:

First, **sensitivity**: The Chow test attains its global unique maximum ($F = 118,074$) precisely at the critical temperature T_c . The analytic mathematical proof confirms that the detection deviation is strictly zero in the analytic sense—the numerical residual deviation on the order of 10^{-8} is entirely explained by the machine-precision limit of finite-precision floating-point arithmetic.

Second, **specificity**: An exhaustive full-parameter search (3000 temperature points $\times$ 4 explained variables), after excluding the known region of $T_c \pm 0.1$, discovered no rupture of comparable magnitude, with a signal-to-noise ratio reaching up to 164,000:1.

Third, **information duality**: The Chow F is rigorously proven to be the information proxy of the thermodynamic order parameter—the F value maps strictly monotonically to the order parameter, spanning seven orders of magnitude. The interaction-effect incremental R^2 is an exact proxy for the strength of a second-order phase transition—attaining a peak value of 0.9992 at T_c .

This calibration provides threefold support for the methodological deployment in this paper. First, the sensitivity and specificity of the Chow test have been independently verified—ruling out the alternative explanation that "the tool itself produces false positives." Second, the statistical output of the Chow test has been proven to be an exact information proxy for physical regime switches—providing a methodological foundation for the leap from statistical discovery to physical interpretation in nuclear physics. Third, the inherent amplification effect of the Chow test on regime-switch signals, based on the quadratic structure of the residual sum of squares, has been algebraically proven—providing a mathematical framework for understanding the extreme magnitude of $\eta > 10^6$ in nuclear physics.

It must be candidly acknowledged that there is a complexity gap between the 2D Ising calibration and the nuclear-physics deployment—the former is the exact solution of a two-state spin system, and the latter is experimental data from a many-body quantum system. The calibration proves the reliability of the tool on a simple system with known truth, but it is not a sufficient condition for the tool to be equally accurate on a

complex system. However, it constitutes a necessary-condition proof of the tool's reliability—if the tool were already unreliable on the simple system, it would be even less likely to be reliable on the complex system.

3. Analysis Results

3.1 Search Completeness: Exhaustive Search Confirms the Statistical Irreplaceability of Liquid-Drop Model Core Variables

Exhaustive testing of all 1- to 3-factor combinations was performed on 1368 complete-data nuclides. **Table 2** reports the optimal execution factor combinations under different VIF constraints.

Table 2: Optimal Execution Factor Combinations from Exhaustive Testing

VIF Constraint	Optimal Combination	Adjusted R ²	Max VIF
VIF < 5	mass_excess_keV + beta_decay_keV + N_over_Z	0.8052	4.71
VIF < 10	mass_excess_keV + Q2B_keV + N_over_Z	0.8481	7.30

The three factors of the VIF < 5 optimal combination—mass excess (overall binding strength), beta-decay energy (decay-channel characteristic), and neutron-to-proton ratio (nuclide-composition asymmetry)—are exactly functionally equivalent to the core terms of the liquid-drop model (volume term, symmetry-energy term) and the decay-channel probe of the shell correction. The proposers of the liquid-drop model (Bethe & Weizsäcker, 1935–1936) found these key variables through physical intuition, whereas the operating system of the Factor Hierarchy Law independently confirmed the statistical irreplaceability of these variables in describing binding energy through exhaustive testing, without any presupposition of nuclear physics theory.

Bayesian model averaging confirmed the global optimal model with overwhelming weight. **Table 3** reports the weight distribution of the top five BMA models.

Table 3: BMA Top-5 Model Weight Distribution

Rank	Factor Combination	BIC	Δ BIC	Weight
1	mass_excess_keV + Q2B_keV + N_over_Z	13574.2	0	1.0000
2	mass_excess_keV + QB_keV + N_over_Z	13879.0	304.8	0.0000
3	mass_excess_keV + beta_decay_keV + N_over_Z	13915.0	340.8	0.0000
4	mass_excess_keV + Qa_keV + N_over_Z	14020.5	446.3	0.0000
5	mass_excess_keV + S2n_keV + Q2B_keV	14033.1	458.9	0.0000

The BIC weight of the optimal model is 1.0000 under double-precision floating-point arithmetic, and the weights of the second-place model and all subsequent models are exactly zero. The BIC gap of 304.8 far exceeds the "decisive evidence" threshold of 10. The BMA factor inclusion probabilities—the cumulative weights of mass_excess_keV, Q2B_keV, and N_over_Z in the top 20 models are 1.0000, 1.0000, and 1.0000, respectively—confirm that these three factors are the irreplaceable core variables in the data.

Candid disclosure: The BMA weight of 1.0000 holds under the current candidate factor pool; the possibility that the weight distribution could change if new factors (such as excitation energy or deformation parameters) are added in the future is not excluded.

Table 4 further confirms the consistency of the four search tools. `mass_excess_keV` ranks first in exhaustive search, Lasso, random forest, and mutual information. `Qa_keV` (alpha decay energy) ranks in the top three in Lasso, random forest, and mutual information but does not enter the top five in the linear exhaustive search—the importance of decay-channel factors leaps under nonlinear structures. `N_over_Z` ranks second in the linear exhaustive search and fourth in the random forest—the neutron-to-proton ratio is a core factor in both linear and nonlinear frameworks.

Table 4: Consistency Comparison of Four Search Tools

Rank	Exhaustive (TOP5 entry count)	Lasso (coefficient)	Random Forest (importance)	Mutual Information (score)
1	<code>mass_excess_keV</code>	<code>mass_excess_keV</code>	<code>mass_excess_keV</code>	<code>mass_excess_keV</code>
2	<code>N_over_Z</code>	<code>Qa_keV</code>	<code>Qa_keV</code>	<code>atomic_mass_micro_u</code>
3	<code>Q2B_keV</code>	<code>N_minus_Z</code>	<code>atomic_mass_micro_u</code>	<code>Qa_keV</code>
4	<code>beta_decay_keV</code>	<code>A_minus_2Z</code>	<code>N_over_Z</code>	<code>N_minus_Z</code>
5	<code>QB_keV</code>	—	<code>N_minus_Z</code>	<code>A_minus_2Z</code>

Integrating the results of the four tools—exhaustive search, Lasso, random forest, and mutual information—the selected execution factors are `mass_excess_keV` + `beta_decay_keV` + `N_over_Z` (optimal under $VIF < 5$, striking a balance between fitting accuracy and collinearity control).

3.2 Modulation Relationship: The η Hierarchy and Regime Factor Determination

This section is divided into three parts: §3.2.1 reports the interaction effect tests and the η hierarchy, establishing the statistical criterion for regime factors; §3.2.2 reports the threshold interaction model and the discovery of $N=25$; §3.2.3 deploys endogeneity exclusion and robustness checks.

3.2.1 Statistical Criterion for Regime Factors

Table 5 reports the interaction effect test results for the 10 candidate regime factors.

Table 5: Interaction Effect Tests for Regime Factors

Regime Factor	Interaction F	Interaction p	Direct Contribution ΔR^2	Judgment
N	565.30	$< 10^{-250}$	< 0.001	Regime Factor
Z	692.94	$< 10^{-250}$	< 0.001	Regime Factor
A	636.07	$< 10^{-250}$	< 0.001	Regime Factor
<code>is_magic_N</code>	22.68	1.28×10^{-12}	< 0.001	Regime Factor

Regime Factor	Interaction F	Interaction p	Direct Contribution ΔR^2	Judgment
is_magic_Z	11.20	2.01×10^{-6}	0.0006	Regime Factor
is_double_magic	—	—	<0.001	Collinear, untestable
dist_to_magic_N	107.04	$< 10^{-23}$	—	Regime Factor
dist_to_magic_Z	118.94	$< 10^{-23}$	—	Regime Factor
N_minus_Z	234.07	$< 10^{-140}$	—	Regime Factor
parity	15.72	3.64×10^{-7}	—	Regime Factor

The conditional independence test further reveals: after adding is_magic_N to the execution factor model, $\Delta R^2 < 0.001$ (displayed as 0.0000 to four decimal places, with a true value on the order of 0.00003); for is_double_magic, $\Delta R^2 < 0.001$; for is_magic_Z, $\Delta R^2 = 0.0006$.

The η hierarchy: As a robustness check, we constructed the continuous shell-correction energy (the residual of the liquid-drop model) as an alternative proxy variable and repeated the interaction effect test. Combined with the main analysis in the body text, this paper has actually measured the η values of three different variables:

1. **$\eta > 10^6$ for continuous N/Z variables:** For N and Z as continuous variables, their interaction effects have $F = 565\text{--}693$, with a direct contribution $\Delta R^2 < 0.001$. The function of continuous N/Z is realized almost entirely through interaction effects—this is the core statistical evidence that "N and Z behave as regime factors under continuous-variable proxies." It must be candidly acknowledged that the $\eta > 10^6$ for continuous N/Z here is estimated from the independent information of the interaction F values of N and Z and their direct contribution ΔR^2 , rather than being calculated through the rigorous ANOVA decomposition as was done for the binary-marker $\eta = 19.19$ and the continuous shell-correction $\eta \approx 0$. The precision of the three differs, but they point to the same conclusion.

2. **$\eta = 19.19$ for the binary magic-number marker:** For is_magic_N, the direct effect $\Delta R^2 = 0.000827$, and the interaction effect $\Delta R^2 = 0.015873$. Even when the regime-factor function of N/Z is compressed into the crudest binary switch, the interaction effect still exceeds the direct effect by more than an order of magnitude.

3. **$\eta \approx 0$ for the continuous shell-correction energy:** The direct effect of the shell-correction energy is $\Delta R^2 = 0.343$, and its interaction effect is $\Delta R^2 = 0.000000$. The influence of the shell-correction energy on binding energy is primarily direct—it is an execution factor, not a regime factor. It must be candidly acknowledged that the shell-correction energy is extracted from the residual of the liquid-drop model—it has already had the smooth trend of N and Z removed. The part of independent particle motion related to the smooth background may have already been absorbed into the liquid-drop model parameters, and the shell-correction energy captures only the fluctuating part of the shell effect. Therefore, $\eta \approx 0$ may partly reflect the effect of the detrending treatment—a complete assessment of the effect of independent particle motion may need to combine both the liquid-drop model parameters and the shell-correction energy.

The current η hierarchy depends on the choice of three specific proxy variables. Continuous N/Z captures the global function of N and Z across the entire N-axis, the binary magic-number marker captures the discrete regime-switch effect at the magic-number positions, and the continuous shell-correction energy captures the difference between independent particle motion and the liquid-drop model trend. Different proxy variables capture different facets of the same physical phenomenon, and their η magnitudes can differ enormously. The conclusion that "N and Z behave as regime factors under continuous-variable proxies" holds under the continuous N/Z proxy, but this determination itself is proxy-variable-dependent—this is a methodological problem not yet fully resolved in the nuclear-physics deployment of the Factor Hierarchy Law. This proxy-variable-dependence problem is, in principle, solvable: the information duality (§2.4) guarantees that the Chow F is the information proxy of some physical order parameter—the η -value differences between proxy variables reflect their different efficiencies in capturing this order parameter. Searching for the proxy variable that maximizes η is equivalent to searching for the optimal statistical representation of the regime-switch order parameter in the nuclear shell. This is one of the directions for subsequent papers.

Physical implications of the η hierarchy: The above three-tier η hierarchy points to a single conclusion—N and Z behave as regime factors under continuous-variable proxies ($\eta > 10^6$), the shell correction is an execution factor ($\eta \approx 0$), and independent particle motion leaves clear, detectable statistical traces in the binding-energy data (direct effect of the shell-correction energy $\Delta R^2 = 0.343$), but it does not itself modulate the exposure coefficients of the execution factors. Independent particle motion exists, but it is not the rule-maker. The rule-makers are N and Z themselves.

Based on the above findings, **Theorem 5 (Nuclear-Physics Regime-Factor Determination Theorem)** is refined:

With the binary magic-number marker as the proxy variable,

$$\eta_X = \frac{\Delta R_{\text{interaction}}^2(X)}{\Delta R_{\text{direct}}^2(X)} > 10^6$$

When the binary magic-number marker is used as the proxy variable, η_X exceeds 10^6 ; the interaction effect of X plays a decisive role in modulating the exposure coefficients of the execution factors, while its direct explanatory power is negligible. When the continuous shell-correction energy is used as an alternative proxy variable, its function is primarily direct ($\eta \approx 0$). The binary marker captures the discrete regime-switch effect at magic-number positions, and the continuous shell-correction energy captures the continuous shell effect across the entire N-axis—the two are different statistical manifestations of the same physical phenomenon under different proxy variables.

Candid disclosure: $\eta > 10^6$ is a threshold specific to nuclear physics—it originates from the extremely high signal-to-noise ratio of nuclear binding-energy data (approximately 1000:1) and depends on the choice of this specific proxy variable, the binary magic-number marker. Part of the contribution to $\eta > 10^6$ may come from the informational advantage of continuous N/Z variables—continuous variables naturally have higher degrees of

freedom for interaction effects and lower direct effects than binary variables. On the exact solution of the 2D Ising model, the 38-fold amplification effect of the Chow F has been rigorously proven to originate from the algebraic structure $F \propto M^2$ (Tang, 2026k). The extreme magnitude of $\eta > 10^6$ in nuclear physics may arise from a homologous mathematical structure—the Chow test, based on the quadratic form of the residual sum of squares, produces amplification effects of different forms in different statistical dimensions. $\eta > 10^6$ should not be regarded as a cross-disciplinarily universal threshold for regime-factor determination but rather as a statistical fingerprint specific to the binary magic-number marker in nuclear physics.

Further methodological support comes from the calibration on the 2D Ising exact solution (§2.4): the interaction effect test in that calibration precisely confirmed a "rule-reset-type" switch ($p = 0.000$, incremental $R^2 = 0.991$), with an exact information duality with the Ehrenfest second-order phase-transition classification. This means: when the same interaction effect test detects, in nuclear physics data, that the interaction effects of N and Z ($F = 565\text{--}693$) vastly exceed their direct effects ($\Delta R^2 < 0.001$), this statistical signature is unlikely to be an artifact of the method itself—the method has already been proven, on a system with known truth, to be capable of precisely distinguishing regime factors from execution factors.

3.2.2 Threshold Interaction and N=25

The threshold interaction model searched for the optimal threshold on the N-axis and locked onto N=25 (coefficient-difference norm = 3361.79). N=25 has no special marking whatsoever in traditional nuclear physics—it is not a magic number, nor a subshell closure, and traditional theory considers it entirely ordinary. However, its physical position lies exactly in the intermediate zone between the two traditional magic numbers N=20 and N=28, the key region where the sd shell transitions from half-filled to fully filled. At N=25, the 1d5/2 orbital is exactly filled (6 neutrons) and the 2s1/2 orbital begins to fill (2 neutrons)—this is a subshell boundary that is the only filling transition within the sd shell that involves a change in orbital angular momentum (from the $l=2$ 1d orbital to the $l=0$ 2s orbital), where the deformation degrees of freedom of the nuclear structure undergo a significant change. The drastic difference in exposure coefficients means that the binding-energy determination equations on either side of N=25 are qualitatively different.

3.2.3 Endogeneity Exclusion and Robustness Checks

To exclude the alternative explanation of endogeneity of $N_{\text{over}}Z$, the instrumental variable method was deployed—using mass number A as an instrument for $N_{\text{over}}Z$. **Table 6** reports the coefficient comparison between OLS and GLS.

Table 6: GLS Coefficient Comparison (OLS vs. GLS)

Variable	OLS Coefficient	GLS Coefficient	Difference
const	10529.46	5862.21	−4667.25
mass_excess_keV	−0.0049	−0.0068	−0.0019

Variable	OLS Coefficient	GLS Coefficient	Difference
beta_decay_keV	0.0443	-0.0019	-0.0462
N_over_Z	-1844.77	+1360.69	+3205.46

The sign of the N_over_Z coefficient reverses between OLS and GLS—from -1844.77 to +1360.69. This means that, after correctly accounting for the covariance structure of the measurement errors, the direction of the contribution of N_over_Z to binding energy undergoes a fundamental change. The sign reversal is evidence for the exclusion of endogeneity and also independent support for the existence of regime switches.

The Chow test on the liquid-drop model residuals further excludes the alternative explanation of "trend artifacts": using the liquid-drop model residual (experimental binding energy minus liquid-drop model prediction) as an alternative explained variable, the Chow test was re-executed. At all 15 key breakpoints, the ratios of the Chow F of the liquid-drop model residual to the Chow F of the original binding energy are all close to 1.00 (range 0.999–1.001). After removing the macroscopic trend of the liquid-drop model, the regime-switch signal strength is almost completely unchanged. This excludes the alternative explanation that "the statistical signal of the Chow test arises from the smooth variation of the liquid-drop model parameters with N."

It must be candidly acknowledged that the extremely high Chow F in the light-nucleus region (N=14–25) (F = 1488–1891) may partly arise from the nonlinear variation of the symmetry-energy term (proportional to $(N-Z)^2/A$) near the N=Z line—this is a known macroscopic effect of the liquid-drop model, not a shell-effect regime switch. The liquid-drop model residual Chow test confirms the authenticity of the signal but does not fully distinguish the relative contributions of "nonlinear symmetry-energy variation" and "shell-effect regime switch" in this region. This distinction requires independent verification on binding-energy data from shell-model calculations.

3.3 Activation-Operator Stratification and the N=14 Statistical Rupture

The exhaustive scanning results of the seven activation operators are shown in **Table 7**.

Table 7: Exhaustive Scanning Results for Seven Activation Operators

Operator	Optimal Breakpoint N	Chow F	Group Affiliation
inv	14	415.27	inv family
log	14	572.24	inv family
inv_sq	14	121.96	inv family
raw	32	1238.13	raw family
sqrt	30	1089.60	raw family
sq	32	1233.44	raw family
cube	32	1217.53	raw family

The operators automatically stratify into two groups—the inv family all precisely lock onto $N=14$ ($F = 121\text{--}572$), and the raw family all lock onto $N=30\text{--}32$ ($F = 1089\text{--}1238$). No operator chooses $N=20$ or $N=28$ as its optimal breakpoint. The Chow F of the inv family at $N=30\text{--}32$ is far below its peak, and the Chow F of the raw family at $N=14$ is similarly far below its peak—the operator selectivity exhibits a sensitivity difference even stronger than the $\text{Width}/W \times M^2$ stratification in particle physics.

This stratification behavior is structurally isomorphic to that in particle physics where Width focuses on the charm region and $W \times M^2$ focuses on the bottom region (Tang, 2026n)—in both cases, different operators lock onto different regime centers. Activation operators are not signal amplifiers but regime selectors—different operators illuminate different segments of the N -axis, corresponding to regime-switch regions dominated by different physical mechanisms. The inv transformation amplifies the variance difference in the light-nucleus region; the inv-family operators are highly sensitive to regime switches in the $N < 20$ region and unanimously lock onto $N=14$. The raw-family operators are sensitive to global variance differences and lock onto $N=30\text{--}32$ —the endpoint of the transition zone of the $N=28$ shell closure. $N=20$ and $N=28$ are the quantum-mechanical starting points of shell closures but are not the statistical centers of gravity of the transition zones. Bai-Perron, aiming to minimize the global residual sum of squares, automatically places the breakpoint at the optimal partition point within the transition zone.

The mathematical root of activation-operator stratification can be understood from the χ -blindness phenomenon in the 2D Ising calibration (Tang, 2026k). On the 2D Ising exact solution, the Chow F of χ is only 16.8 (completely silent), but the Chow F of $1/\chi$ explodes to $\sim 10^{21}$ —different mathematical transformations alter the alignment between the explained variable and the sensitivity structure of the Chow test. The stratification behavior in nuclear physics—the inv family locking onto $N=14$ and the raw family locking onto $N=30\text{--}32$ —may follow the same mathematical logic: the inv transformation amplifies the variance difference near the $N=Z$ line in the light-nucleus region, causing it to resonate with the sensitivity structure of the Chow test at $N=14$; the raw transformation preserves the global variance structure, causing it to produce the maximal response at the center of gravity of the transition zone at $N=30\text{--}32$. This explanation is heuristic—a complete mathematical proof requires further research.

It needs to be clarified that activation-operator stratification (inv family $\rightarrow N=14$, raw family $\rightarrow N=30\text{--}32$) and the Tool-Dependence Theorem (Tang, 2026m) describe statistical phenomena at two different levels. The Tool-Dependence Theorem describes how the detection results of different tools near the same regime-switch center are distributed within a narrow band—for example, PELT at $N=23$ and kernel change-point at $N=23$, but both are within the transition zone of $N=20\text{--}32$, converging on the same regime center. In activation-operator stratification, the inv family locks onto $N=14$ and the raw family locks onto $N=30\text{--}32$ —these two positions are approximately 17 neutrons apart, far beyond the narrow-band range described by the Tool-Dependence Theorem. The former is the multi-tool convergence of the same regime switch; the latter is the operator

selectivity of different regime centers. Both fall under the category of "tools are variable," but they correspond to different levels of regime-switch structure.

3.3.1 Triple Cross-Validation and Supplementary Evidence for the N=14 Statistical Rupture

N=14 has no marking whatsoever in traditional nuclear physics—it is not a magic number, nor a subshell closure, and traditional theory considers it ordinary. Yet three independent tools unanimously lock onto this position, supplemented by one additional piece of evidence:

First line: The three activation operators of the inv family (inv, log, inv_sq) all lock their optimal breakpoint at N=14. Three different mathematical transformations independently converge on the same position—this is a powerful guarantee of the signal's authenticity.

Second line: The threshold interaction model, searching for the optimal threshold on the N-axis, locks onto N=25 as the position of maximal coefficient difference. Within the transition zone from N=14 to N=25, the exposure coefficients of the binding-energy equation are undergoing drastic adjustment—N=14 is the starting point of this adjustment.

Third line: Among 14 tested boundaries, the PICO intervention effect attains its maximal intercept jump at N=14 (−245.98 keV, $p < 0.001$)—the baseline of the binding-energy equation undergoes the most drastic jump when crossing N=14. The PICO framework treats the regime switch as an "intervention" and the intercept jump as the "intervention effect"—this quantification method is methodologically completely isomorphic to the quantification of the density jump at valence=2 in chemistry (Tang, 2026t).

Supplementary evidence: All eight breakpoints of the kernel change-point cosine kernel concentrate at $N < 15$ —among the four kernel functions, the cosine kernel has the highest sensitivity to regime switches in the light-nucleus region, and its eight breakpoints are densely distributed in the $N = 4-14$ interval, constituting an independent cross-validation with the lock-on by the inv family of activation operators (Tang, 2026w).

Based on the above triple cross-validation and supplementary evidence, **Theorem 4 (N=14 Statistical Rupture Theorem)** is refined:

$$F_{\text{precursor}}(N = 14) = \max_{T \in \{14\}} \|\Delta\beta_{\text{inv}}(N = T)\|$$

where $\Delta\beta$ is the difference vector of the exposure coefficients of the execution factors on either side of breakpoint T, and the inv-family operators have the highest sensitivity to its variation in the light-nucleus region. A statistically significant structural rupture exists at N=14 (confirmed by triple independent tool cross-validation, supported by one supplementary piece of evidence).

Distinction between statistical fact and physical interpretation: The existence of a statistically significant structural rupture at N=14—this is a statistical fact confirmed by triple independent tool cross-validation. Interpreting this rupture as "the earliest precursor warning signal of the N=20 shell closure" is a cross-disciplinary physical hypothesis built upon the precursor-cascade regularities independently verified by

the Factor Hierarchy Law in particle physics and chemistry. Precursor cascades have been verified in three independent disciplines—in particle physics, strange $T_2 \approx$ charm T_1 and charm $T_2 =$ bottom precursor (Tang, 2026o); in chemistry, the early warning of the first ionization energy at period = 2.0 (Tang, 2026s); and in statistical physics, the M_{sp}/U_4 double-peak fission of the 3D Ising model (Tang, 2026l). These independent verifications constitute a cross-disciplinary regularity discovery: in regime-switch systems, precursor signals tend to appear at the starting point of the transition zone before the main switch point. $N=14$ lies exactly at the starting point of the transition zone between $N=8$ (p-shell) and $N=20$ (sd-shell)—this position is consistent with the statistical signature of precursor cascades. This cross-disciplinary analogy provides not confirmatory evidence but heuristic support—independent verification within nuclear physics remains indispensable.

$N=14$ lies between the traditional magic numbers $N=8$ and $N=20$, the key region where the light-nucleus region transitions from the p-shell to the sd-shell. At $N=14$, the $1d_{5/2}$ orbital begins to fill, and the nuclear structure transitions from the p-shell ($l=1$ dominated) to the sd-shell ($l=0,2$ mixed). This transition involves a change in orbital angular momentum—this is precisely the physical root of the regime switch. In chemistry, the regime-switch intensity $F(l) \propto l^2$ (Tang, 2026s)—the change in orbital angular momentum at $N=14$ produces a detectable structural rupture in the variance-partition space.

The magnitude of the Chow F at $N=14$ (415–572) can be referentially compared with that in the 2D Ising calibration (118,074). Although the physical natures of the two systems are completely different, the Chow test exhibits extremely high sensitivity to regime switches in both—in 2D Ising, the F value erupts to 118,074 at T_c , and in nuclear physics, the inv transformation amplifies the F value from a weak signal in the raw metric to 415–572. This cross-system sensitivity consistency further supports the statistical authenticity of the structural rupture at $N=14$.

Known experimental anomalies exist for nuclides near $N=14$ —the β -decay half-life of ^{14}C (5730 years) is approximately 1000 times longer than predicted by the shell model, the first excited 0^+ state of ^{16}O is anomalously low (6.05 MeV), and the magnetic moment of ^{17}O deviates from the Schmidt line by about 30%. These known anomalies obtain a possible qualitative unified explanation within the regime-switch framework: they are the inevitable manifestations of the interactive modulation of multiple sets of rules at the $N=14$ precursor boundary. It must be candidly acknowledged that these known anomalies are currently only qualitatively consistent and have not yet been independently verified by the Chow test on excitation-energy spectrum data.

Specific verification paths and falsification conditions:

- **Verification path (a):** Scan the energies of all known excited states in the $N=10$ – 18 interval in NUBASE2020 using the Chow test. Falsification condition: if no significant Chow peak appears at $N=14$ ($p > 0.05$), then the $N=14$ statistical rupture may be confined to the ground-state binding energy and does not constitute a cross-energy-state regime-switch signal.

- **Verification path (b):** Test whether the neutron separation energy S_n exhibits an anomalous jump at $N=14$ —anomaly defined as the jump amplitude exceeding the 95th percentile of S_n jump amplitudes at each neutron number in the $N=10$ – 18 interval. The test must be performed separately at different Z values (e.g., $Z = 8, 10, 12, 14, 16, 18, 20$) to confirm whether the anomaly consistently appears across multiple Z values. Falsification condition: if the S_n jump amplitude at $N=14$ does not exceed this threshold, or if the anomaly appears only for a few Z values, then the $N=14$ rupture may not be a Z -independent regime boundary.

- **Verification path (c):** Perform a Bai-Perron double-breakpoint blind test in the $N=10$ – 18 interval. Falsification condition: if the algorithm does not detect a breakpoint at $N \approx 14$ (± 2 neutrons), then the $N=14$ rupture may be only a single-breakpoint structure and does not satisfy the double-breakpoint characteristic of a precursor cascade.

3.4 Structural Breakpoints: Bai-Perron Optimal 10 Breakpoints from Full-N-Axis Blind Test

The Bai-Perron sequential blind test is the core tool of the structural breakpoint dimension. The algorithm, completely ignorant of any nuclear physics concepts, automatically selects the optimal number and locations of breakpoints by exhaustively searching the global residual sum of squares and introducing a breakpoint penalty.

Table 8 reports the BIC selection under different maximum breakpoint counts.

Table 8: Full-N-Axis Bai-Perron Multi-Breakpoint BIC Selection

max_bkps	BIC	Breakpoint Positions (N)
3	14723.6	[83, 104, 129]
5	13831.4	[29, 65, 83, 104, 129]
7	13859.7	[29, 65, 83, 104, 118, 129, 145]
10	13793.7	[15, 29, 65, 83, 104, 118, 129, 137, 145, 150]
15	13848.7	[15, 29, 44, 54, 65, 83, 90, 104, 112, 118, 129, 137, 145, 150, 156]

The BIC attains its minimum value of 13793.7 at $\text{max_bkps} = 10$ —the algorithm automatically selected 10 regime-switch points as the optimal solution. These 10 breakpoints—[15, 29, 65, 83, 104, 118, 129, 137, 145, 150]—cover the entire nuclide chart from $N=15$ to $N=150$.

It must be candidly acknowledged that the gap between BIC at $\text{max_bkps} = 7$ (13859.7) and at $\text{max_bkps} = 10$ (13793.7) is 66.0—while far above the "strong evidence" threshold of 10, this gap is not enormous relative to the absolute magnitude of the BIC (approximately 0.5%). This means there may be a slight model-selection uncertainty in the optimal number of breakpoints between 7 and 10. The coexistence of the 10-breakpoint and 7-breakpoint solutions itself suggests that some regime-switch signals are secondary structures resolved only at higher breakpoint counts. The conclusions of Theorems 1–3 do not depend on the precise choice of the number

of breakpoints—whether 7 or 10 breakpoints, the traditional magic numbers $N=20$ and $N=50$ are not selected, and $N=28$, $N=82$, and $N=126$ are all systematically shifted.

Table 9 systematically compares these 10 breakpoints with the traditional magic numbers.

Table 9: Comparison of Bai-Perron 10 Breakpoints with Traditional Magic Numbers

Traditional Magic Number	Closest Blind Breakpoint	Deviation (neutrons)	Selected in Optimal Model?
$N=20$	15	−5	No
$N=28$	29	+1	Yes
$N=50$	65	+15	No
$N=82$	83	+1	Yes
$N=126$	129	+3	Yes
—	104, 118, 137, 145, 150	—	No corresponding traditional magic number

$N=20$ and $N=50$ are not selected in the optimal model—the algorithm determined, under completely blind conditions, that these two traditional magic numbers are not the centers of gravity of the regime switches of the binding-energy equation. $N=28$ is shifted by +1 to $N=29$, $N=82$ is shifted by +1 to $N=83$, and $N=126$ is shifted by +3 to $N=129$. $N=104$, 118, 137, 145, and 150 have no correspondence whatsoever in the traditional magic-number table.

To further quantify the statistical status of each traditional magic number, the individual ΔBIC values were calculated for the five magic numbers—comparing the BIC difference of the model with and without that magic number as a breakpoint. **Table 10** reports the results.

Table 10: Individual ΔBIC Values and Classification of the Five Traditional Magic Numbers

Magic Number N_0	BIC with	BIC without	ΔBIC	Offset $ \Delta $	Classification
$N=20$	34765.7	34251.3	−514.4	5	Single-particle-gap type
$N=28$	34261.8	34677.7	+415.9	1	Deformation-transition type
$N=50$	34102.8	34251.3	+148.5	15	Single-particle-gap type
$N=82$	34248.6	34260.5	+11.9	1	Deformation-transition type
$N=126$	34251.3	34219.0	−32.3	3	Single-particle-gap type

For $N=20$, $\Delta\text{BIC} = -514.4$ —forcing $N=20$ into the breakpoint set increases the BIC by 514.4, significantly worsening the model fit. For $N=28$, $\Delta\text{BIC} = +415.9$ —including the $N=28$ breakpoint reduces the BIC by 415.9, overwhelming statistical evidence. For $N=82$, $\Delta\text{BIC} = +11.9$ —a positive contribution but of minimal magnitude, a marginal improvement. For $N=126$, $\Delta\text{BIC} = -32.3$ —including the $N=126$ breakpoint actually worsens the fit. The separate analysis of $N=126$ further shows that, within the $N=120$ –135 interval, the BIC values at five consecutive N values are exactly the same (all 34251.3)—Bai-Perron faces a flat BIC surface in the heavy-nucleus region, a statistical manifestation of decreasing sample size and signal-to-noise ratio.

Although $N=82$ meets the criterion for deformation-transition type with $\Delta\text{BIC} = +11.9$, its evidential strength is far weaker than that of $N=28$ ($\Delta\text{BIC} = +415.9$, a difference of approximately 35 times). This magnitude difference suggests that there may be significant intensity grades within the deformation-transition type—the shell closure at $N=82$ may involve weaker deformation coupling than that at $N=28$, or may mix with other mechanisms. The current binary classification system compresses this internal intensity difference into a single criterion, failing to adequately reflect the difference in their statistical characteristics.

Although $N=50$ and $N=20$ are both classified as single-particle-gap type, their statistical manifestations are starkly different. The $\Delta\text{BIC} = -514.4$ for $N=20$ means that forcing $N=20$ into the breakpoint set severely deteriorates the model fit— $N=20$ leaves almost no statistical trace in the binding-energy equation. The $\Delta\text{BIC} = +148.5$ for $N=50$ means that a statistical signal does exist at $N=50$, but the optimal partition point lies far away at $N=65$ —its transition zone is extremely wide (15 neutrons). This difference suggests that the single-particle-gap type may internally divide into two subtypes: the $N=20$ type (no statistical signal) and the $N=50$ type (has a broad signal but with an excessive offset). The current binary classification merges the two, failing to reflect this potential difference.

It needs to be clarified that $\Delta\text{BIC} > 0$ means that the breakpoint makes a positive contribution when forcibly included—including this breakpoint improves the model fit. However, Bai-Perron, in its global optimization, considers the joint effect of the entire breakpoint set and may choose a different, superior combination of breakpoints, resulting in this breakpoint not being selected in the optimal model. The $\Delta\text{BIC} = +148.5$ for $N=50$ is a positive contribution, but its offset is too large ($|\Delta| = 15$), and Bai-Perron determined that $N=65$ can more effectively cover the regime-switch region of $N=50$. The improvement magnitude of $\Delta\text{BIC} = +11.9$ for $N=82$ is minimal (far below the decisive threshold of 10), and Bai-Perron determined that it is not worth the added model complexity. These are two different statistical issues— ΔBIC reflects "the contribution when included alone," and the Bai-Perron optimal breakpoint set reflects "the choice under global joint optimization."

Based on the above findings, three quantitative theorems are refined:

Theorem 1 (Magic-Number Shift Theorem):

$$N^* = N_0 + \Delta(N_0)$$

The offset Δ is determined by the difference between the statistical center of gravity of the shell-closure transition zone and its quantum-mechanical starting point. When the shell closure involves only a single-particle energy gap, $|\Delta| \geq 5$, and the breakpoint may be completely excluded from the optimal model (e.g., $N_0 = 20, 50$); even if $\Delta\text{BIC} > 0$ (e.g., $\Delta\text{BIC} = +148.5$ for $N_0 = 50$), an excessive offset prevents its entry into the optimal model. When the shell closure also involves a deformation transition, $|\Delta| \leq 3$, and the breakpoint is retained but with a shifted position (e.g., $N_0 = 28 \rightarrow N^* = 29$; $N_0 = 82 \rightarrow N^* = 83$).

Theorem 2 (Statistical Distinction Theorem of Shell-Closure Physical Mechanisms):

$$\Delta\text{BIC}(N_0) = \text{BIC}_{\text{without}} - \text{BIC}_{\text{with}}$$

When $\Delta\text{BIC}(N_0) > 0$ and $|\Delta| \leq 3$, the shell closure involves a deformation transition (e.g., $N_0 = 28, 82$); when $\Delta\text{BIC}(N_0) \leq 0$ or $|\Delta| \geq 5$, the shell closure involves only a single-particle energy gap (e.g., $N_0 = 20, 50, 126$). All five traditional magic numbers conform to this classification criterion. Significant intensity grades exist within the deformation-transition type—the $\Delta\text{BIC} = +415.9$ of $N=28$ (strong deformation-transition type) and the $\Delta\text{BIC} = +11.9$ of $N=82$ (weak deformation-transition type) differ by approximately 35 times. This difference may arise from three factors: (1) $N=28$ involves a large deformation transition from the sd to the pf shell, while $N=82$ involves a milder deformation transition from the light lanthanides to the heavy lanthanides; (2) the signal-to-noise ratio of data in the heavy-nucleus region near $N=82$ is lower than that in the medium-mass region near $N=28$; (3) ΔBIC is only one dimension of statistical distinction, and the local magnitude of the Chow F may reveal further differences. Differences also exist within the single-particle-gap type—the statistical manifestations of $N=20$ ($\Delta\text{BIC} = -514.4$, no statistical signal) and $N=50$ ($\Delta\text{BIC} = +148.5$, has a signal but with an excessive offset) are starkly different. The current binary classification merges the two, failing to reflect this potential difference.

Theorem 3 (Magic-Number Node Theorem): In the one-dimensional N-axis projection,

$$N_0^{\text{magic}} = \text{geometric node} \neq \text{statistical center of gravity} = N^*$$

Traditional magic numbers are not fundamental constants of nuclear physics but rather the geometric nodes of the shell-closure transition zones. The shell model describes the quantum-mechanical behavior of single particles in a mean field—magic numbers are the nodes of the microscopic quantum structure. The Factor Hierarchy Law describes the statistical structure of the many-nucleon system—the blind breakpoints are the centers of gravity of the macroscopic statistical structure. The non-coincidence of the two is not a theoretical conflict but the manifestation of the respective validity of two physical levels.

Inductive basis and universality of Theorems 1–3: The current classification criterion is an inductive regularity based on the five traditional magic numbers. The universality of a regularity constituted by five data points—whether it applies to newly discovered shell closures in the superheavy-nucleus region or near the neutron drip line—requires further verification on independent data. As a falsifiable prediction: if a new shell closure (e.g., $N=184$) is discovered in the superheavy-nucleus region in the future, Theorem 1 predicts that the optimal breakpoint of Bai-Perron will be shifted to $N \approx 184 + \Delta$ ($\Delta \geq 0$), and Theorem 2 predicts that the sign of its ΔBIC and $|\Delta|$ will determine whether it belongs to the deformation-transition type or the single-particle-gap type. If the offset and ΔBIC pattern of the new shell closure are completely different from those of the five traditional magic numbers, the inductive basis of Theorems 1–3 will be weakened.

The independent full-Z-axis Bai-Perron blind test confirms a completely isomorphic shift behavior. At $\text{max_bkps} = 5$, $\text{BIC} = 36594.2$, with breakpoints $Z = [12, 24, 59, 75, 90]$; the global optimum is at $\text{max_bkps} = 15$ with $\text{BIC} = 36213.8$. The traditional proton magic numbers $Z = 8, 20, 28, 50, 82$ —not a single one is

precisely locked onto. The N-axis and Z-axis independent blind tests produce completely independent regime-switch networks, yet their structural characteristics are completely isomorphic.

The SupF full-N-axis scan attains its maximum Chow $F = 1238.13$ at $N=32$, with $p < 10^{-16}$ — $N=32$ is also not a traditional magic number. The CUSUM test yields a maximum deviation of 6.07, far exceeding the 5% critical value, confirming that the equation parameters are globally unstable on the N-axis.

Candid disclosure: The CUSUM critical value displayed as 0.00 is a statsmodels version-compatibility issue and does not affect the qualitative conclusion that the maximum deviation of 6.07 far exceeds any reasonable critical value. The calculations of the Chow test and Bai-Perron test are based on standard formulas, and their results do not depend on the specific version implementation of statsmodels.

3.5 N-Z Two-Dimensional Chow: One-Dimensionally Shifted Magic Numbers Relocked in Two Dimensions

The N-Z two-dimensional Chow scan searches for the optimal four-quadrant partition boundary in the (N, Z) plane. **Table 11** reports the Top 5 results.

Table 11: N-Z Two-Dimensional Chow Test Top 5

Rank	N Breakpoint	Z Breakpoint	Chow F
1	20	14	524.24
2	20	20	508.88
3	28	20	420.71
4	29	20	410.41
5	28	27	401.21

The optimal two-dimensional partition is at $N=20/Z=14$ ($F = 524.24$), slightly higher than $N=20/Z=20$ ($F = 508.88$). $N=20$ is not a local extremum on the one-dimensional Chow F curve (the Chow F of $N=19$, 1252.75, is higher than the 1231.59 of $N=20$), yet in the two-dimensional plane, the intersection of $N=20$ and $Z=14$ is locked as the strongest signal. A magic number shifted in one dimension is relocked in two dimensions—the regime switch exhibits a clearer structure in higher dimensions.

$Z=14$ (silicon) is not a traditional proton magic number, yet it appears at the strongest two-dimensional Chow position—this means that the two-dimensional regime-switch center of the $N=20$ shell closure is not at the position of the traditional doubly magic nucleus ^{40}Ca ($N=20$, $Z=20$) but in the lighter, proton-rich region (near $N=20$, $Z=14$). It must be candidly acknowledged that this finding is a post-hoc observation rather than an a priori prediction, and its physical meaning requires further investigation.

$Z=27$ (cobalt) appears in the two-dimensional Chow Top 5—nuclides near it are known in nuclear-structure research to exhibit complex shape-coexistence phenomena. This is also a post-hoc observation.

Higher-resolution two-dimensional Chow scans (e.g., with a step size of 0.5) can test whether $Z=14$ and $Z=27$ still maintain significance in finer scans—this is deferred to subsequent research.

Based on the above findings, **Theorem 6 (Dimension-Recovery Theorem)** is refined:

$$\max_{T_N, T_Z} F_{\text{Chow}}(T_N, T_Z) > \max_{T_N} F_{\text{Chow}}(T_N), \quad T_N = N_0^{\text{magic}}$$

A traditional magic number shifted or excluded on the one-dimensional N-axis may be relocked as the strongest regime-switch signal on the two-dimensional (N, Z) plane. Dimensional elevation can recover regime-switch structures lost in the one-dimensional projection. Theorem 3, "magic numbers are geometric nodes, not statistical centers of gravity," applies to the one-dimensional projection—in the two-dimensional plane, a magic-number pair may recover its status as a statistical center of gravity as a component of the intersection. This dimensional dependence is an inevitable consequence of the projection of a regime switch in a multi-variable space.

3.6 Negative Controls: Three Lines of Defense Excluding Statistical Artifacts

To ensure that all findings are not statistical artifacts or artifacts driven by measurement errors, three independent lines of defense were deployed, progressing from the weakest to the strongest alternative explanation. **Table 12** reports the summary results.

Table 12: Summary of Three Lines of Negative-Control Defense

Line of Defense	Test Content	Core Result	Alternative Explanation Excluded
Permutation Test	500 random shuffles of N labels, recomputing optimal Chow F	Permutation F median = 2.14, empirical p = 0.002 (empirical lower bound of 500 permutations)	Probability of random noise producing a breakpoint of comparable magnitude $\leq 0.2\%$
Fake-Data Control	Replacing binding energy with random noise, executing the full Chow test	Adjusted $R^2 = 0.0007$ (expectation ≈ 0), all interaction terms non-significant (p = 0.235–0.926)	Method does not produce false positives on structureless data
WLS Residual Chow	Using AME2020 official measurement errors as weights, performing a Chow scan on the weighted residuals	Optimal breakpoint $N=17$, Chow F = 482.55, $p < 10^{-16}$	Regime switch is not driven by drip-line nuclides with large measurement errors

The three lines of defense jointly exclude statistical artifacts and alternative explanations driven by measurement errors. The optimal breakpoint of the WLS residual Chow test, $N=17$, is close to but slightly shifted from the Bai-Perron optimal breakpoint $N=15$ —quantitative analysis shows that the nuclides with the

highest WLS weights are all concentrated in the heavy-nucleus region (mean $N = 141.1$), with zero nuclides near $N=17$. The shapes of the Chow F curves for WLS and OLS in the $N=14$ – 20 interval are highly similar (the WLS/OLS ratio monotonically decreases from 0.86 at $N=14$ to 0.72 at $N=18$), confirming that the regime-switch signal is not affected by the choice of measurement-error weights. The WLS breakpoint $N=17$ is a natural consequence of the global weight structure and does not constitute an alternative to or challenge against the Bai-Perron breakpoint $N=15$.

The Chow test on the liquid-drop model residuals has already been reported in §3.2.3—at all 15 key breakpoints, the ratios of the residual Chow F to the original Chow F are all close to 1.00, excluding the alternative explanation of "trend artifacts."

The cross-regime rupture of reaction-energy errors provides an independent verification at the measurement-error level: all seven reaction-energy error columns ($S2n_err_keV$, $S2p_err_keV$, Sn_err_keV , Sp_err_keV , Qa_err_keV , QB_err_keV , $Q2B_err_keV$) exhibit systematic high/low ratio changes at all nine regime boundaries. At $N=126$, the ratios for the seven columns, 1.68–1.86, all have $p < 0.001$. This independent signal further excludes the alternative explanation that "the regime switch is a statistical artifact of the binding-energy data analysis"—even in reaction-energy error data with completely different explained variables, the structural jump at the regime boundary still exists.

Candid disclosures: (1) $p = 0.002$ is the empirical lower bound of 500 permutations—if increased to 5000 permutations, the p -value could further shrink to 0.0002 or lower; (2) the strong inherent structure of nuclear binding-energy data (traditional magic numbers, shell effects, etc.) renders the permutation test statistically conservative—randomly shuffling N labels does not completely destroy the shell structure in the data, because the shuffled data still retain the physical correlation between N and Z ; (3) the $\eta > 10^6$ for continuous N/Z does not depend on the precise calculation of the p -value and does not require the assumption of independent and identically distributed residuals—it is a more robust statistical criterion for regime-factor determination. However, the analysis of the continuous shell-correction energy shows that $\eta > 10^6$ depends on the choice of proxy variable—the binary marker captures the discrete magic-number effect, and the continuous shell-correction energy captures the continuous shell effect. The $\eta \approx 0$ for the shell-correction energy may partly reflect the effect of the detrending treatment—a complete assessment of the effect of independent particle motion may need to combine both the liquid-drop model parameters and the shell-correction energy; (4) the CUSUM critical value displayed as 0.00 is a statsmodels version-compatibility issue and does not affect the qualitative conclusion.

4. Discussion

4.1 Physical Implications and Interconnections of the Six Theorems

The six theorems refined in this paper jointly sketch a complete picture of the regime switch of the nuclear shell from four dimensions.

Theorems 1–3 (the Magic-Number Shift Theorem, the Statistical Distinction Theorem of Shell-Closure Physical Mechanisms, and the Magic-Number Node Theorem) jointly accomplish the repositioning of traditional magic numbers within the statistical structure—magic numbers are not fundamental constants but the geometric nodes of the shell-closure transition zones. The individual ΔBIC values of all five traditional magic numbers conform to the classification criterion of Theorem 2. The statistical center of gravity is given by the Bai-Perron blind optimal breakpoints; the non-coincidence of the two is an inevitable manifestation of the regime switch in a finite-nucleus system. Theorem 3 applies to the one-dimensional N-axis projection—in the two-dimensional intersection, certain magic-number pairs can become statistical centers of gravity, which is the content of Theorem 6. The current classification criterion is an inductive regularity based on the five traditional magic numbers, and its universality requires further verification on independent data.

Theorem 4 (the N=14 Statistical Rupture Theorem) establishes the existence of a statistically significant structural rupture at N=14—confirmed by triple independent tool cross-validation and further supported by one supplementary piece of evidence. The physical interpretation of this rupture (as the earliest precursor warning signal of the N=20 shell closure) is a physical hypothesis based on cross-disciplinary pattern recognition and requires independent experimental testing.

Theorem 5 (the Nuclear-Physics Regime-Factor Determination Theorem) reveals the η hierarchy—continuous N/Z $\eta > 10^6$, binary magic-number marker $\eta = 19.19$, continuous shell-correction energy $\eta \approx 0$. The three-tier η hierarchy points to a single conclusion: N and Z behave as regime factors under continuous-variable proxies ($\eta > 10^6$), the shell correction is an execution factor ($\eta \approx 0$). Independent particle motion leaves clear, detectable statistical traces in the binding-energy data (direct effect of the shell-correction energy $\Delta R^2 = 0.343$), but it does not itself modulate the exposure coefficients of the execution factors. Independent particle motion exists, but it is not the rule-maker. The rule-makers are N and Z themselves. $\eta > 10^6$ is a threshold specific to nuclear physics, dependent on the choice of this particular proxy variable, the binary magic-number marker, and requires recalibration in other disciplines.

Theorem 6 (the Dimension-Recovery Theorem) reveals that the regime switch exhibits a clearer structure in higher dimensions. Theorems 3 and 6 jointly describe the complete picture of magic numbers in statistical space: they are geometric nodes in the one-dimensional projection and statistical centers of gravity in the two-dimensional intersection.

Argument-strength grades of the theorem system: The six theorems differ significantly in argument strength and universality. **Table 13** summarizes the argument strength, inductive basis, and universality limitations of each theorem.

Table 13: Argument Strength and Inductive Basis of the Six Theorems

Theorem	Argument Strength	Inductive Basis	Universality Limitation
Theorem 1 (Magic-Number Shift)	Moderate (inductive)	Five traditional magic numbers	Requires verification on independent shell closures
Theorem 2 (Statistical Distinction)	Moderate (inductive)	Five traditional magic numbers; $\Delta\text{BIC} = +11.9$ for $N=82$ is marginally significant	Binary classification may oversimplify
Theorem 3 (Magic-Number Node)	Strong (definitional)	One-dimensional N-axis projection	Some magic-number pairs recover as statistical centers of gravity in the two-dimensional plane
Theorem 4 ($N=14$ Rupture)	Strong for statistical fact, weak for physical interpretation	Single point $N=14$, triple cross-validation	Precursor hypothesis requires independent experiment
Theorem 5 (η Hierarchy)	Strong (mathematical foundation)	Comparison of three proxy variables	η value depends on proxy-variable choice
Theorem 6 (Dimension Recovery)	Moderate (one data point)	$N=20/Z=14$ as primary evidence	Whether it applies to other magic-number pairs requires verification

4.2 Implications for the Axioms of the Liquid-Drop Model and the Shell Model

Exhaustive testing, without any presupposition of nuclear physics theory, confirmed the statistical irreplaceability of the core variables of the liquid-drop model—mass excess (volume term), neutron-to-proton ratio (symmetry-energy term), and decay energy (shell-correction probe)—in describing binding energy (BMA weight 1.0000). This is an independent verification of the physical reasonableness of the liquid-drop model.

The η hierarchy provides a more precise statistical description of the shell-model axiom: independent particle motion leaves clear statistical traces in the binding-energy data—the direct effect of the continuous shell-correction energy $\Delta R^2 = 0.343$, with an interaction effect of zero. The shell correction is the execution-layer manifestation of independent particle motion—it directly contributes to binding energy. But the rule-makers are N and Z themselves—their interaction-to-direct-effect ratio $\eta > 10^6$ means their function is realized almost entirely through modulating the exposure coefficients of the execution factors. Independent particle

motion exists, but it is not the fundamental organizing principle—it is one of the execution mechanisms regulated by the regime factors N and Z .

This paper strictly distinguishes the argument boundaries of two propositions:

- **Proposition A (completed in this paper):** The η hierarchy reveals that, under continuous-variable proxies, N and Z behave as regime factors and the shell correction is an execution factor; the Bai-Perron blind test proves that magic numbers are not optimal partition points in the statistical sense; independent particle motion exists but is not the rule-maker.

- **Proposition B (deferred to subsequent papers, Tang, 2026w, 2026x):** Independent particle motion is not the fundamental organizing principle of the nuclear shell—this requires the proof that the regime-switch network systematically outperforms the shell model in completeness, self-consistency, and explanatory power (three lines of evidence: 15% coverage, liquid-drop masking effect, shell-correction compensation traces), jointly supported by the methodological-priority argument.

The task of this paper is to provide the first link—it is the most fundamental link, because if η were of the same order of magnitude as the direct effect, and if Bai-Perron had precisely locked onto all traditional magic numbers, then the entire chain of argument would have no reason to unfold.

4.3 Methodological Contributions and Limitations

The core methodological contributions of this paper lie in: deploying the operating system of the Factor Hierarchy Law to the domain of nuclear physics for the first time; exhaustive testing confirming the statistical irreplaceability of the core variables of the liquid-drop model with zero physical presuppositions; the Bai-Perron blind test automatically identifying 10 regime-switch points completely ignorant of the concept of magic numbers; activation-operator stratification revealing statistical ruptures undetectable by traditional methods; $N=14$ being confirmed as a statistically significant structural rupture by triple independent tool cross-validation. The Chow test on the liquid-drop model residuals confirmed that the regime-switch signal is not an artifact of the liquid-drop model trend, and the η hierarchy revealed the sensitivity of regime-factor function to the choice of proxy variable.

Candid disclosures:

1. The analysis of this paper is entirely based on ground-state nuclide binding-energy data (AME2020); ground-state binding energy only reflects the static properties of nuclides at absolute zero temperature. The regime audit of excitation-energy spectra will be undertaken in subsequent papers;
2. The full- Z -axis Bai-Perron analysis still requires finer-step scanning;
3. A statistically significant structural rupture exists at $N=14$ —this is a statistical fact confirmed by triple independent tool cross-validation. Interpreting it as "the earliest precursor warning signal of the $N=20$ shell

closure" is a physical hypothesis based on cross-disciplinary pattern recognition and requires independent experimental testing (for specific verification paths and falsification conditions, see §3.3.1);

4. The $\Delta R^2 < 0.001$ for N and Z is a result displayed to four decimal places, with a true value on the order of 0.00003, and is not an exact mathematical zero—however, this does not alter the qualitative conclusion that $\eta > 10^6$;

5. The BMA weight of 1.0000 is a result under the current candidate factor pool; the possibility that the weight distribution could change if new factors are added in the future is not excluded;

6. The $\eta > 10^6$ for continuous N/Z does not depend on the precise calculation of the p-value and does not require the assumption of independent and identically distributed residuals. However, the analysis of the continuous shell-correction energy shows that $\eta > 10^6$ depends on the choice of proxy variable—the binary marker captures the discrete magic-number effect, and the continuous shell-correction energy captures the continuous shell effect. The $\eta \approx 0$ for the shell-correction energy may partly reflect the effect of the detrending treatment—a complete assessment of the effect of independent particle motion may need to combine both the liquid-drop model parameters and the shell-correction energy;

7. The CUSUM critical value displayed as 0.00 is a statsmodels version-compatibility issue and does not affect the qualitative conclusion;

8. The extremely high Chow F in the light-nucleus region (N=14–25) may partly arise from the nonlinear variation of the symmetry-energy term—the liquid-drop model residual Chow test confirmed the authenticity of the signal but did not fully distinguish the relative contributions of "nonlinear symmetry-energy variation" and "shell-effect regime switch";

9. The statements of Theorems 1–3 are operationalized statistical correspondences— Δ is the numerical difference between the Bai-Perron optimal breakpoint and the traditional magic number. This difference is an objective statistical fact, but the physical meaning of the "shift" requires further verification on independent data. The current classification criterion is an inductive regularity based on the five traditional magic numbers, and its universality requires further verification on independent data;

10. The $\Delta BIC = -32.3$ and $|\Delta| = 3$ for N=126 require further verification on a larger superheavy-nuclide dataset—within the N=120–135 interval, the BIC surface is almost flat, and Bai-Perron's breakpoint selection faces statistical uncertainty in this region.

5. Conclusion

This paper deployed the Factor Hierarchy Law and Testability Norms to perform a zero-preset regime audit on the full AME2020 nuclide dataset. Eighteen candidate execution factors were exhaustively tested, and

four search tools unanimously confirmed the optimal execution factor combination. The core findings are distilled into four tiers of empirical discovery and six quantitative theorems:

Tier 1—The η hierarchy and regime-factor determination (Theorem 5): Continuous N/Z $\eta > 10^6$, binary magic-number marker $\eta = 19.19$, continuous shell-correction energy $\eta \approx 0$. The three-tier η hierarchy points to a single conclusion: N and Z behave as regime factors under continuous-variable proxies ($\eta > 10^6$), the shell correction is an execution factor ($\eta \approx 0$). Independent particle motion leaves clear, detectable statistical traces in the binding-energy data (direct effect of the shell-correction energy $\Delta R^2 = 0.343$), but it does not itself modulate the exposure coefficients of the execution factors. Independent particle motion exists, but it is not the rule-maker. The rule-makers are N and Z themselves.

Tier 2—Activation-operator stratification and the $N=14$ statistical rupture (Theorem 4): The inv family all lock onto $N=14$, and the raw family all lock onto $N=30-32$. $N=14$ is confirmed as a statistically significant structural rupture by triple independent tool cross-validation, with one supplementary piece of evidence providing further support—this is a statistical fact. Interpreting it as "the earliest precursor warning signal of the $N=20$ shell closure" is a physical hypothesis based on cross-disciplinary pattern recognition and requires independent experimental testing. Three falsifiable verification paths are given in the body text.

Tier 3—Bai-Perron optimal 10 breakpoints from blind testing (Theorems 1–3): An optimal 10 breakpoints are locked with overwhelming statistical evidence ($BIC = 13793.7$), covering $N=15-150$. Traditional magic numbers are systematically shifted; $N=20$ and $N=50$ are not selected in the optimal model. The individual ΔBIC values of all five traditional magic numbers conform to the Statistical Distinction Theorem of Shell-Closure Physical Mechanisms. The Magic-Number Shift Theorem, the Statistical Distinction Theorem, and the Magic-Number Node Theorem establish the repositioning of traditional magic numbers within the statistical structure.

Tier 4— $N-Z$ two-dimensional strongest partition (Theorem 6) and three lines of negative control: $F = 524.24$ at $N=20/Z=14$ —a magic number shifted in one dimension is relocked in two dimensions. Three lines of negative control (permutation test $p = 0.002$, fake-data $R^2 = 0.0007$, WLS residual Chow $F = 482.55$) jointly exclude statistical artifacts.

This paper establishes the statistical existence of regime-switch signals in the nuclear shell—continuous N/Z $\eta > 10^6$, binary magic-number marker $\eta = 19.19$, and the fact that magic numbers are not optimal partition points in the statistical sense. These two statistical findings constitute the necessary premises for the subsequent argument that "independent particle motion is not the fundamental organizing principle of the nuclear shell." This paper does not claim to have completed the full argument from statistical discovery to physical proposition—the closure of the latter requires the joint support of the three lines of evidence in subsequent papers (15% coverage, liquid-drop masking effect, shell-correction compensation traces) and the methodological-priority argument (Tang, 2026w, 2026x).

Data Availability Statement

All data in this study come from the international Atomic Mass Evaluation AME2020 and the nuclear properties database NUBASE2020, obtained through the official channel of the IAEA Atomic Mass Data Center (AMDC). The data are completely public, and any researcher can independently reproduce all results. The master data table `nuclear_master_table.csv` and the analysis code are available upon request.

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